ENTERPRISE COMPUTING

Advanced Java Architecture

From JDBC & Servlet fundamentals to cloud-native Spring Boot microservices.

Jakarta EE · Spring Ecosystem · Distributed Systems

DATA PERSISTENCE & ORM

JDBC & High-Throughput Storage



Type-4 Driver Architecture

Direct native-protocol database connectivity.

PreparedStatement & Security

Precompiled execution preventing SQL injection.

HikariCP Connection Pool

Zero-overhead thread pool scaling.

WEB LAYER EXECUTION

Servlet Request–Response Lifecycle

PHASE 01

init()

One-time servlet loading and configuration initialization.

PHASE 02

Filter Pipeline

Pre-processing authentication and request transformation.

PHASE 03

service()

Concurrent dispatch to doGet and doPost handlers.

PHASE 04

destroy()

Graceful teardown and thread resource cleanup.

MODERN ECOSYSTEM

Spring Boot & Cloud Infrastructure

Inversion of Control (IoC)

Decoupled dependencies via automated runtime injection.

Spring Data JPA & ORM

Repository abstraction over raw Hibernate queries.

Cloud-Ready Microservices

Embedded Netty/Tomcat runtime with declarative APIs.





Ready for Enterprise Java

Build scalable, secure, and distributed applications using the modern Java stack.

Jakarta EE • Spring Boot • REST • Microservices

Image Sources



https://b2287888.assetcdn.net/2.0/2287888/wp-content/uploads/2025/10/modern-data-center-server-racks.jpg?lossy=1&strip=1&webp=1

Source: www.presentationgo.com



https://understanding.mit.edu/sites/default/files/images/helping-data-centers.jpg

Source: news.mit.edu